Rishabh Kalwar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.13
Bakshis Springdales Sr sec School	Class XII (cbse)	2024	86.00%
Bakshis Springdales Sr sec School	Class X (cbse)	2022	91.20%

PROJECTS

CPU Cache Simulator | GITHUB LINK

- **Enabled performance comparison across cache architectures** by designing and implementing a configurable simulator supporting Direct-Mapped, Set-Associative, and Fully Associative designs
- **Improved extensibility for future replacement policies** by architecting a modular, object-oriented system implementing LRU, FIFO, LFU, and Random eviction strategies
- **Quantified memory-system performance trade-offs** by simulating memory access traces and computing hit rate, miss rate, AMAT, and eviction statistics across configurations
- **Made performance trends interpretable** by building visualization tools to compare cache behavior across varying sizes, block sizes, and associativity levels

Simple Distributed Load Balancer | GITHUB LINK

- **Built** a **multithreaded C++ load balancer** using **POSIX sockets** to route client requests across multiple backend servers via **round-robin scheduling**.
- **Implemented TCP health checks** with **automatic traffic rerouting**, enabling **fault-tolerant failover** when backend nodes become unavailable.
- **Engineered thread-per-connection concurrency** with **atomic operations** for **thread-safe request distribution** across simultaneous clients.

Credit-risk-model | GITHUB LINK

- **Reduced modeled manual underwriting workload by 75%** by designing a 3-tier risk-decision system (auto-approve/manual-review/auto-reject) that used percentile-based cutoffs to isolate low-risk from high-risk applicants
- **Achieved 0.75 AUC-ROC in default prediction** by building an end-to-end ML pipeline in **Python/Scikit-learn**, training and comparing Logistic Regression and Random Forest classifiers on applicant financial and behavioral data
- **Delivered a deployable, business-ready pipeline** by engineering feature importance analysis, model evaluation, and business-impact quantification alongside the core model, translating raw predictions into an operational decision framework .

DNS Security & Threat Intelligence Filtering | GITHUB LINK

- **Hardened** a Pi-hole DNS resolver with **DNSSEC validation** to prevent **DNS spoofing and cache-poisoning attack**s.
- **Integrated live URLhaus/abuse.ch threat** feeds to **block malicious domains** and malware distribution at the DNS layer.
- Resolved **DNS routing and blocklist migration** issues during deployment, ensuring reliable network-level filtering.

TCP Client-Server Chat Application | GITHUB LINK

- Developed a **TCP client-server messaging system** in C supporting **real-time bidirectional communication** over sockets.
- Implemented connection lifecycle management using **socket(), bind(), listen(), accept(), and connect()** APIs.
- Improved communication reliability with **input buffering, read/write error handling, and graceful socket termination.**

SKILLS

Languages: C/C++, Python, R, Bash .

Technical Skills : Data Structures and Algorithm, OOPS, System Design, Networking .

Libraries : Pandas, NumPy, Matplotlib, Os, Scikit-Learn, Seaborn **.**

Networking & Concurrency:sys/socket.h, arpa/inet.h, thread, mutex, atomic .

RELEVANT COURSES

CSE : Data Structures & Applications, Introduction to Programming, Artificial Intelligence.

Mathematics : Probability & Random Variables, Calculus, Linear Algebra, Differential Equations, Biostatistics.

Computational Biology : Big Data Biology & Biological Databases, Machine Learning in Bioinformatics, R & Python for Biologists.

POSITIONS OF RESPONSIBILITY

Internship Coordinator, Office of Career Services (OCS), IIT Hyderabad

Core Member, Rang De Manch (RDM), IIT Hyderabad

EXTRACURRICULAR

- Represented IIT Hyderabad at the **Inter IIT Cultural Meet** as a member of the Rang De Manch theatre contingent.
- Enjoy running, playing basketball, watching Formula 1, NFL, and participating in online strategy games. Watch classic Hollywood films, sci-fi shows, and anime (big fan of One Piece).
- Volunteered with the **National Service Scheme(NSS)** at IIT Hyderabad, contributing to community service and social outreach initiatives.